Yiyan Liao

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EDUCATION

School of Life Sciences, Peking University Sep. 2023 – Jun. 2027 (Expected)

Bachelor of Engineering in Bioinformatics

- Ranking: 4/133; GPA: 3.75/4.00
- Selected Coursework: Linux-based Essential Bioinformatics (A); Innovative Experiments in Bioinformatics (A); Bioinformatics (A); Artificial Intelligence for Protein Design (94/100); Computational Systems Biology (93/100)

SELECTED PUBLICATIONS

†Equal contribution. *Corresponding author.

[1] **MGTbind: a comprehensive database of molecular glue ternary interactome**
Jintao Zhu†, ***Yiyan Liao***†, Haoyu Lin†, ..., Luhua Lai*, Jianfeng Pei*
Nucleic Acids Res., 54(D1), D1500–D1509, 2026. https://doi.org/10.1093/nar/gkaf1075

[2] **Benchmarking Cofolding Methods for Molecular Glue Ternary Structure Prediction**
Yiyan Liao†, Jintao Zhu†,*, Juan Xie, Luhua Lai*, Jianfeng Pei*
J. Chem. Inf. Model., 65(20), 11136–11148, 2025. https://doi.org/10.1021/acs.jcim.5c01860

[3] **Molexar: A Unified Multimodal Molecular Foundation Model for Drug Design**
Haoyu Lin†, ***Yiyan Liao***†, Jinmei Pan, Xinliao Ling, Luhua Lai*, Jianfeng Pei*
arXiv preprint, 2026. https://doi.org/10.48550/arXiv.2606.25865

[4] **MLIPDiff: Guiding Denoising Diffusion Models for Structure-based Drug Design via Machine-learned Interatomic Potentials**
Yiyan Liao†, Amin Tavakoli†, Yu Zhang, Anima Anandkumar*
2nd SIMBIOCHEM Workshop at NeurIPS 2026. https://doi.org/10.48550/arXiv.2610.33315

RESEARCH EXPERIENCE

Anima AI + Science Lab, California Institute of Technology Pasadena, CA

Summer Undergraduate Research Fellow (SURF) — Advisor: Prof. Anima Anandkumar Mar. 2026 – Present

MLIPDiff: Physics-Guided Molecular Generation (NeurIPS SIMBIOCHEM 2026; ongoing)

- Guided diffusion models with machine-learned interatomic potential forces trained on quantum-mechanical data, eliminating protein-ligand clashes and improving predicted docking scores.
- Fine-tuned TargetDiff with DDPO using a UMA-computed interaction energy reward, achieving similar gains.
- Ongoing: extending the framework to autoregressive (Pocket2Mol) and flow-matching (FLOWR.ROOT) models.

Molecular Design Laboratory, Peking University Beijing, China

Undergraduate Researcher — Advisors: Dr. Jianfeng Pei and Prof. Luhua Lai Sep. 2024 – Present

MGTbind: a Comprehensive Database of Molecular Glue Ternary Interactome (Co-first author, Nucleic Acids Res.)

- Curated 3,093 molecular glue ternary complex entries from 227 publications, covering structures and activities.
- Modeled 3,607 complexes with AlphaFold 3 and annotated physicochemical descriptors and interactions.

Benchmarking Cofolding Methods for Molecular Glue Ternary Structure Prediction (First author, J. Chem. Inf. Model.)

- Built MGBench, a molecular glue ternary complex benchmark, from 221 curated structures with time-based splits isolating 88 unseen targets.
- Benchmarked five SOTA cofolding models including AlphaFold 3, showing their successes stem from memorization rather than atomic-level rules, with the largest gaps on novel E3 ligase systems.

DiffTernary: A Diffusion Model for Molecular Glue Ternary Complex Docking (Ongoing)

- Developed an SE(3)-equivariant diffusion model that predicts ternary docking poses by refining rigid-body placement and ligand torsions.
- Improved the acceptable protein-interface prediction rate by 10% over current SOTA DeepTernary (DockQ).

Molexar: A Unified Multimodal Molecular Foundation Model for Drug Design (Co-first author, preprint)

- Co-developed the Fragment-SELFIES molecular language and a Gemma 2-based autoregressive decoder conditioned on properties, pharmacophores, sequences, and binding pockets.

- Led evaluation against 20+ baselines, attaining SOTA results on CrossDocked2020 test set and MolGenBench.
- Led candidate generation and screening for DPP4, ACO1, ACOD1, and GPR6, compounds are in activity assays.

DegronCraft: Computational Design of Molecular Glue-Dependent Degrons (Ongoing)

- Developed a multi-stage design pipeline using BoltzGen, ESM3, Boltz-2, and SCISOR, yielding 27-aa degron candidates with improved predicted interface energies; undergoing experimental validation.

Peking University International Genetically Engineered Machine (iGEM) Team Beijing, China

Wet Lab Leader — Advisor: Dr. Long Qian Jan. 2024 – Oct. 2024

General Disease Detection with Protease and Aptamer (2024 iGEM Gold Medal)

- Tested thrombin-binding aptamers with SPR/EMSA, performed mutagenesis and click chemistry conjugation.
- Directed wet-lab strategy for a 15-member team and presented the final project defense at the Grand Jamboree.

PROFESSIONAL EXPERIENCE

Tencent Beijing, China

Biological Large Language Model Intern Jun. 2025 – Dec. 2025

- Developed a six-stage multi-agent system with retrieval-augmented generation and NCBI tools for automated *de novo* plasmid design, raising benchmark accuracy from 10% to 51%.

Moonshot AI Beijing, China

Large Language Model Reinforcement Learning Intern Jan. 2025 – Jun. 2025

- Built biology-domain reinforcement-learning data and ran pilot experiments for Kimi K2 on the Graduate-Level Google-Proof Q&A (GPQA) benchmark, informing the full-scale training runs.

SELECTED HONORS & AWARDS

National Scholarship for Undergraduate Students (top 4% of the cohort) 2026

Caltech Summer Undergraduate Research Fellowship ($8,110 Grant) 2026

School of Life Sciences 87 Excellent Research Scholarship, First Prize (1st out of 10 finalists) 2026

Peking University Merit Student (top 10% of the cohort) 2025, 2026

Beijing Natural Science Foundation Undergraduate Research Program ($7,000 Grant) 2025

International Genetically Engineered Machine (iGEM) 2024 Gold Medal (Wet Lab Leader) 2024

34th International Biology Olympiad (IBO) Gold Medal (3rd out of 293) 2023

ACADEMIC ACTIVITIES & SERVICE

Teaching Assistant of Biochemistry Laboratory, Peking University Sep. 2025 – Sep. 2026

- Instructed 25 students per semester in weekly biochemistry laboratory sessions.
- Established an *in silico de novo* design workflow for the GST enzyme using the protein language model ESM3.
- Awarded the Outstanding Teaching Assistant Award, School of Life Sciences, Peking University (Jun. 2026).

Student Advisor of Peking University iGEM Team 2025 Jan. 2025 – Oct. 2025

- Selected members and guided research for Team Peking 2025, which won a Global Top 10 ranking.

Member of Undergraduate Honors Program in Biology (UHPB), Peking University Sep. 2024 – Present

- Gave 4 journal club talks (1 hour each) and attended 16 more, once presenting to the senior author of the paper.
- Delivered English talks on the molecular glue computational landscape (2025) and physics-guided molecular generation (2026, Best Poster Presenter Award) at UHPB Annual Symposia, to 100+ peers and 10 professors, including international faculty.

SKILLS & OTHER INFO

- **Computational Methods & Tools:** Python, PyTorch, PyTorch Geometric, R, Bash, Git, Linux, SLURM, Schrödinger, GROMACS, Rosetta, AutoDock Vina, AlphaFold 3, ESM3, LLaMA Factory, PyMOL, ChimeraX
- **Experimental Methods:** Molecular Cloning, Site-Directed Mutagenesis, Protein Expression and Purification, ÄKTA HPLC, SPR, EMSA, SDS-PAGE, Western Blot, Co-IP, PCR/qPCR, Bacterial Culture
- **Languages:** TOEFL: 103/120